

6 (a) $\text{p.d.} = \frac{\text{work done} / \text{energy transformed}}{\text{charge}}$ (from electrical to other forms)

(b) (i) maximum 20 V

(ii) minimum = $(600 / 1000) \times 20$
= 12 V

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- (c) (i) use of $1.2\text{ k}\Omega$ M1
 $1/1200 + 1/600 = 1/R$, $R = 400\ \Omega$ A1 [2]
- (ii) total parallel resistance ($R_2 + \text{LDR}$) is less than R_2 M1
(minimum) p.d. is reduced A1 [2]